Lukas José Ferrer

ML & AI SYSTEMS ENGINEER

lukasjferrer@gmail.com | 310.801.8751 | Los Angeles, CA

linkedin.com/in/lukasferrer | github.com/Ljferrer

LinkedIn

ML engineer building clinical AI, self-hosted inference, and developer tools. Shipped systems processing **100M+ medical-record pages**; creator of WAR, an open-source platform for parallel AI coding with independent review and verified integration.

FEATURED OPEN SOURCE

WAR | Work Audit Refine

Creator & Maintainer | June 2026 - Present

- Built a Claude Code plugin that schedules task DAGs, runs parallel agents in isolated Git worktrees, and integrates approved changes through a serial merge queue.
- Designed commit-pinned reviews by 1-5 independent auditors, severity-based blocking, and test/lint gates; added SQLite FTS5 memory and Git-based crash recovery.

Explore WAR

Watch demo

EXPERIENCE

Lead ML & DevOps Engineer | Sequoia Port Inc.

Mar 2021 - Present | Remote / Los Angeles, CA

- **Led the migration of medical-record Activation to self-hosted Qwen 3.6 (27B), bringing measured AI pipeline cost below $0.01/page**: $0.009628 across 110.1K page activations (Oct 21-26, 2026; excluding chat costs).
- Built asynchronous LLM routing across detection, transcription, extraction, classification, and summarization using Step Functions, SQS/SNS, and DynamoDB, with durable recovery, validated outputs, and controlled fallback to paid APIs.
- Moved seven page and summary compute steps to ECS Fargate and ARM64; added Spot capacity with on-demand recovery for bulk page processing.
- Operationalized hybrid GPU inference using local ECS Anywhere workers and queue-driven cloud fallback, with Spot/on-demand capacity that scales to zero when idle.
- Shipped a Python-native clinical record-summary pipeline with pgvector retrieval, parallel extraction, shadow comparisons, transactional persistence, and runtime fallback.
- Built biomedical translation evaluations and extended extraction/deduplication benchmarks with degraded scans, clinical-error reports, and paired, blinded model comparisons.
- Protected physician edits during reprocessing through database-authoritative record reconciliation and idempotent persistence.

Lukas José Ferrer

Production AI / ML infrastructure / Agent orchestration

EXPERIENCE CONTINUED

Sequoia Port Inc. | Selected earlier impact

- Architected a fault-tolerant medical-record NLP pipeline serving Fortune 100 healthcare customers: **100M+ pages processed** and **5M+ pages/month on 2-4 GPUs**.
- Built clinical AI products that reduced insurance claim review time by **65%** and specialist escalations by **40%**; automated segmentation of 1,000+ page cases for physician review in under an hour.
- Improved clinical retrieval from **87% to 95% Recall@5** using a domain ontology and graph database; built multi-stage case summaries with citation validation and PHI anonymization.

Red Team (Contract) | OpenAI

Jan 2024 - Jun 2024 | Remote

- Red-teamed GPT-4o for calibrated uncertainty in clinical contexts. Designed synthetic-condition evaluations that exposed plausible but nonexistent ICD-10 codes and tested ontology-grounded retrieval.

Regulatory Software Engineer | Neuralink

Oct 2020 - Jan 2021 | Fremont, CA

- Supported FDA regulatory preparation for a Class III implantable brain-computer interface; assessed design-control gaps and proposed remediation aligned with medical-device requirements.

Research Programmer | USC Information Sciences Institute

May 2018 - Oct 2020 | Marina del Rey, CA

- Integrated research Transformer architectures into a neural machine translation training library; trained thousands of models and improved translation quality across low-resource languages.
- Built an ETL pipeline ingesting **100,000+ news articles/day** for the SAGE geopolitical forecasting platform, improving relevant-document retrieval by **73% (Recall@5)**.

TECHNICAL SKILLS

AI & ML: PyTorch, Transformers, multimodal models, LoRA, distillation, Qwen, Anthropic/OpenAI APIs, RAG, pgvector, FAISS, evaluation, citation validation.

Infrastructure: AWS CDK, Lambda, Step Functions, ECS/Fargate, ECS Anywhere, EC2, SQS/SNS, S3, DynamoDB, PostgreSQL, Docker, Linux, GPU serving, CI/CD.

Development: Python, TypeScript/JavaScript, SQL, Bash, React, Ruby on Rails, Claude Code, MCP, multi-agent orchestration, Git worktrees, SQLite FTS5.

Domain: Clinical document processing, PHI protection, HIPAA-oriented pipelines, differential privacy (PATE), medical-device regulatory workflows.

EDUCATION

M.S. Biomedical Engineering | USC Viterbi School of Engineering | Aug 2017

B.S. Biomedical Engineering | USC Viterbi School of Engineering | May 2016